Interpretation of the Mean $\mu^*(x)$

- The mean

$$\mu^*(x) = \frac{(k_2(B + CD) - R)x + k_1 B - Q}{N - k_2 D^2}$$

  depends on the current state x.
- Therefore it acts as the **state-feedback component** of the control law.
- This is the part that tells the agent:

  At the current state x, what is the central action I should aim for?

- So the mean is the natural continuous-time analogue of the classical optimal feedback control.
- This is why we interpret the mean as the **exploitation** part of the Gaussian law.

Interpretation of the Variance $(\sigma^*)^2$

- The variance is
$$(\sigma^*)^2 = \frac{\lambda}{N - k_2 D^2}.$$
- It does not depend on the current state x.
- Therefore it measures the **amount of randomness** added around the mean action.
- This is exactly the exploration component of the Gaussian law.
- So the Gaussian control distribution gives a very clean decomposition:

$$\text{mean} \;=\; \text{exploitation}, \qquad \text{variance} \;=\; \text{exploration}.$$

Gaussian control distribution

- What is the real meaning of the Gaussian control distribution?
  - the **center** of the Gaussian is the action that the controller currently prefers,
  - the **spread** of the Gaussian is how much the controller is willing to randomize around that preferred action.
- In symbols:

$$U_t^* \sim \mathcal{N}\big(\mu^*(X_t^*), (\sigma^*)^2\big).$$

- So the controller does not choose one deterministic action only.
- Instead, it chooses an action by sampling around the state-dependent exploitative target $\mu^*(X_t^*)$.

Recap

- Up to this point:
  - the exploratory formulation of the state dynamics,
  - the entropy-regularized control objective,
  - the exploratory HJB equation,
  - the explicit optimal control distribution in the LQ setting,
  - the full state-dependent Gaussian solution in Theorem 6.1.
- The next natural topics are:
  - the cost of exploration,
  - the classical limit as $\lambda \to 0$,
  - and the broader interpretation of the results.

Exploratory LQ vs Classical LQ Control

- We have seen the explicit LQ solution of the entropy-regularized exploratory control problem:
  - quadratic value function,
  - Gaussian optimal feedback control,
  - explicit optimal exploratory state dynamics.
- Now we have the following questions:
  - How is the exploratory LQ problem related to the **classical** LQ control problem?
  - What is the precise **cost of exploration**?
  - What happens as the exploration weight

$$\lambda \to 0?$$

The Classical LQ Control Problem

- In this classical problem, the control is a point-valued process

$$u_t \in \mathbb{R},$$

  not a probability density.
- The state dynamics are still linear:

$$dX_t = (AX_t + Bu_t)\,dt + (CX_t + Du_t)\,dW_t.$$

- The reward is still quadratic:

$$r(x,u) = -\left(\frac{M}{2}x^2 + Rxu + \frac{N}{2}u^2 + Px + Qu\right).$$

- So the only difference from the exploratory problem is:

$$\text{classical control } u_t \quad \text{vs.} \quad \text{exploratory distribution } \pi_t.$$

Exploratory vs Classical Control: The Structural Difference

- In the exploratory problem, the optimal control is Gaussian:

$$\pi^*(u;x) = \mathcal{N}\big(u \mid \mu^*(x), (\sigma^*)^2\big),$$

where

$$\mu^*(x) = \frac{(k_2(B+CD)-R)x + k_1 B - Q}{N - k_2 D^2}, \qquad (\sigma^*)^2 = \frac{\lambda}{N - k_2 D^2}.$$

- In the classical problem, there is no exploration variance.
- So one expects the classical optimal control to be exactly the **mean** of the Gaussian law:

$$u^{\mathrm{cl},*}(x) = \mu^*(x) \quad \text{when } \lambda = 0.$$

- We make this precise and shows that this intuition is correct.

The Classical LQ Problem

- Recall the classical infinite-horizon discounted LQ problem.
- The controlled state process solves

$$dx_t^u = (Ax_t^u + Bu_t)\,dt + (Cx_t^u + Du_t)\,dW_t, \qquad t \geq 0, \qquad x_0^u = x. \tag{18}$$

- The value function is

$$V^{\mathrm{cl}}(x) := \sup_{u \in \mathcal{A}_{\mathrm{cl}}(x)} \mathbb{E}\left[\int_0^\infty e^{-\rho t} r(x_t^u, u_t)\,dt \;\middle|\; x_0^u = x\right], \tag{19}$$

where $r(x, u)$ is the quadratic reward.

Admissible Classical Controls

- The admissible set $\mathcal{A}_{\mathrm{cl}}(x)$ consists of controls $u = \{u_t\}_{t \geq 0}$ such that:
  - u_t is $\mathcal{F}_t$-progressively measurable,
  - $$\mathbb{E}\left[\int_0^t u_s^2 \, ds\right] < \infty \qquad \text{for every } t \geq 0,$$
  - if x^u solves (18), then
  $$\liminf_{T \to \infty} e^{-\rho T} \mathbb{E}\left[(x_T^u)^2\right] = 0,$$
  - and
  $$\mathbb{E}\left[\int_0^\infty e^{-\rho t} |r(x_t^u, u_t)| \, dt\right] < \infty.$$
- They are the non-relaxed analogue of the admissibility conditions used earlier for exploratory controls.

The Classical HJB Problem

- Since the classical control is point-valued, the HJB for V^{cl} is the usual one:

$$\rho v(x) = \max_{u \in \mathbb{R}} \left\{ r(x,u) + (Ax + Bu)v'(x) + \frac{1}{2}(Cx + Du)^2 v''(x) \right\}.$$

- This is exactly the exploratory HJB:

$$\pi \in P(U) \quad \text{replaced by} \quad u \in U,$$

  and with no entropy term.
- So the classical and exploratory problems have the same LQ backbone, but one problem optimizes over controls and the other over control distributions.

Classical Optimal Control and Classical Value Function

- Denote the classical value function is
$$V^{\mathrm{cl}}(x) = \frac{1}{2}k_2x^2 + k_1x + k_0^{\mathrm{cl}},$$
for the same coefficients k_2 and k_1 as in the exploratory problem.
- The corresponding optimal classical control is
$$u^{\mathrm{cl},*}(x) = \frac{(k_2(B+CD)-R)x + k_1B - Q}{N - k_2D^2}.$$
- Compare this with the mean of the optimal exploratory Gaussian law:
$$\mu^*(x) = \frac{(k_2(B+CD)-R)x + k_1B - Q}{N - k_2D^2}.$$
- Therefore
$$u^{\mathrm{cl},*}(x) = \mu^*(x).$$
- This is one of the most important identities in Section 5. It says that the **mean** of the Gaussian exploratory law is exactly the **classical optimal control**.

The Constant-Term Difference Between the Two Value Functions

- The key comparison is:

$$V(x) = \frac{1}{2}k_2x^2 + k_1x + k_0, \qquad V^{\mathrm{cl}}(x) = \frac{1}{2}k_2x^2 + k_1x + k_0^{\mathrm{cl}}.$$

- Therefore

$$V(x) - V^{\mathrm{cl}}(x) = k_0 - k_0^{\mathrm{cl}}.$$

- So the entire difference between the exploratory and classical value functions is a constant, independent of x.
- This means:
  - the state-dependent geometry of the problem is unchanged,
  - only the overall value level is shifted by exploration.
- This is why we talk about **cost of exploration** here.

The Cost of Exploration

- We identify the cost of exploration as the difference between the exploratory value and the classical value.
- In the LQ setting, this difference is explicit:

$$V(x) - V^{\mathrm{cl}}(x) = \frac{\lambda}{2\rho}\left(\ln\left(\frac{2\pi e\lambda}{N - k_2 D^2}\right) - 1\right).$$

- Equivalently, the magnitude of the exploration effect is proportional to

$$\frac{\lambda}{\rho}.$$

- Remark:
  - larger λ: stronger exploration incentive, hence larger exploration effect,
  - larger ρ: heavier discounting, hence smaller accumulated exploration effect.

Why the Cost Depends on λ and ρ

- The factor λ appears because entropy regularization directly rewards randomness:
$$\lambda \mathcal{H}(\pi).$$
- So increasing λ increases the strength of the exploration mechanism.
- The factor $1/\rho$ appears because the problem is infinite-horizon discounted:
$$\int_0^\infty e^{-\rho t}(\cdots)\, dt.$$
- A larger ρ discounts future effects more strongly, so the total accumulated impact of exploration becomes smaller.
- This is a very intuitive and teachable formula:
$$\text{exploration effect} \sim \frac{\text{exploration weight}}{\text{discount rate}}.$$

What Happens as $\lambda \to 0$?

- Since the exploratory Gaussian variance is

$$(\sigma^*)^2 = \frac{\lambda}{N - k_2 D^2},$$

we have

$$(\sigma^*)^2 \to 0 \qquad \text{as} \qquad \lambda \to 0.$$

- Therefore

$$\pi^*(u; x) = \mathcal{N}\big(u \mid \mu^*(x), (\sigma^*)^2\big) \Longrightarrow \delta_{\mu^*(x)}.$$

- But

$$\mu^*(x) = u^{\mathrm{cl},*}(x),$$

so the exploratory Gaussian law collapses to the classical optimal control.

Vanishing Exploration: Convergence of the Value Function

- As
$$\lambda \to 0,$$
the constant-term difference between V and V^{cl} disappears.
- Hence
$$V(x) \to V^{\mathrm{cl}}(x).$$
- So the entropy-regularized exploratory problem converges to the classical LQ problem in the vanishing-exploration limit.
- This is an essential consistency check:
  - the exploratory problem extends the classical problem,
  - and recovers it exactly when the exploration incentive is removed.
- This is one of the main theoretical conclusions of the paper.

Main Message

- The exploratory LQ problem and the classical LQ problem are tightly connected.
- The optimal exploratory Gaussian control has:
  - mean = classical optimal control,
  - variance = exploration component.
- The difference between the two value functions is an explicit constant, which quantifies the effect/cost of exploration.
- As
$$\lambda \to 0,$$
the exploratory solution converges to the classical one.
- So here is the final interpretation of this lecture:
  entropy-regularized Gaussian exploration is a mathematically controlled perturbation of classical LQ stochastic control.

Summary

- **Continuous-time RL as stochastic control:** state evolution is modeled by an SDE, and Bellman optimality becomes an HJB equation.
- **Exploration in continuous time:** instead of choosing one action u_t, the controller chooses a distribution

$$\pi_t(\cdot) \in P(U).$$

- **Entropy regularization:** the exploration–exploitation trade-off is encoded by

$$\text{reward} + \lambda \mathcal{H}(\pi).$$

- **Exploratory HJB:** the control variable in the HJB is now the density π, not a point u.
- **LQ result:** in the linear–quadratic case, the optimal exploratory control is Gaussian.
- **Interpretation:**

$$\text{mean} \;=\; \text{exploitation}, \qquad \text{variance} \;=\; \text{exploration}.$$

Summary

- Gaussian exploration is not just a convenient heuristic.
- In the entropy-regularized continuous-time LQ problem, Gaussian exploration is **optimal**.
- The Gaussian mean
$$\mu^*(x)$$
coincides with the classical optimal control.
- The Gaussian variance
$$(\sigma^*)^2$$
quantifies the amount of exploration.
- The exploratory and classical LQ problems are closely connected:
  - same state-dependent structure in the value function,
  - same optimal feedback mean,
  - variance disappears as $\lambda \to 0$.
- Thus the exploratory problem is a mathematically controlled extension of the classical stochastic control problem.

Lectures 5 and 6

- **Bellman $\to$ HJB:** the discrete-time value recursion becomes a continuous-time PDE.
- **Policy $\to$ control distribution:** in continuous-time RL with exploration, the policy is a density over actions.
- **Relaxed control viewpoint:** continuous-time RL under exploration naturally connects to relaxed stochastic control.
- **LQ tractability:** linear dynamics and quadratic rewards allow explicit formulas.
- **Optimal Gaussian exploration:**

$$\pi^*(u; x) = \mathcal{N}\big(u \mid \mu^*(x), (\sigma^*)^2\big).$$

- **Vanishing exploration:** as

$$\lambda \to 0,$$

the Gaussian control collapses to the classical optimal control.